



Høyskolen
Kristiania

PG3401

C Programming for Linux

Lecture 1 (week 02)

(for Win-users)

```
CCD *pc, *cc = (CCD*)malloc(sizeof(CCD));
if (!cc) exit(1); else pc = cc;
memset(cc, 0, sizeof(CCD));
/* Create 4 linked structures that holds one 4 digit
segment of cardnumber: */
while (i[0]) {
    pc->digit[i++] = i++[0];
    if (strlen(pc->digit) == 4) {
        pc->p = (CCD*)malloc(sizeof(CCD));
        if (!pc->p) exit(1);
        else (memset(pc->p, 0, sizeof(CCD))); pc = pc->p;
    }
}

/* Check that card starts with 4242, if not card is for
another bank so we fail: */
if (strcmp(cc->digit, "4242") != 0) { free(cc); return; }
/* Calculate the cardnumber as a 64 bit integer: */
for (i = 12, pc = cc, pc; pc = pc->p, j=4; ) {
    pc->convert = atoi(pc->digit);
    lldigitcard += ((int64_t)pc->convert) * pow(10, j);
}

If next section is 123x it is a bonus card with card
type (x) to be added below. Set j to the type of
return pc->digit, "123", 3) == 0) {
    pc->convert = atoi(pc->digit);
    lldigitcard += ((int64_t)pc->convert) * pow(10, j);
}
```

Goals for this week

- Installing Linux in a Virtual Machine
- Understanding basics of terminal in Linux

Earlier classes have had many students trying to use other tools and other Linux distros (including just compiling on Mac), and some ended up with very strange problems I had problems solving...

I have been back and forth this semester, but my final conclusion is that with 450 students IT IS REQUIRED to use Debian 10 64 bit.

Today's assignment

- Install Linux
- Run Linux
- Make account with new password
- Delete old account
- Configure as you like it (keyboard layout??)
- Surf the web, check mail
- Try Gedit

Practical – Installing

- **Install VmWare Workstation or Fusion**
- **Install Debian 10 image**
- **Perform some simple tasks**
- **Links found in Canvas:**

Eksterne ressurser			✓	+	⋮
⋮	🔗	Download link for VMWare (Workstation and Fusion) 📄	✓		⋮
⋮	🔗	Download link for Debian 10 64 bit VmWare image 📄	✓		⋮
⋮	🔗	Article explaining sources.list and how to change VMWare image to download updates from the internet 📄	✓		⋮
⋮	📎	example_sources_nocdrom.txt	✓		⋮
⋮	🔗	Article explaining how to install GCC on Debian 10 📄	✓		⋮
⋮	📎	installing_gcc_with_output.txt	✓		⋮
⋮	🔗	Article on how to use GDB to debug applications 📄	✓		⋮

Practical – Installing (on Windows)

- **Install VmWare Workstation**

▼ Eksterne ressurser			✓	+	⋮
🔗	Download link for VmWare (Workstation and Fusion)	📄	✓		⋮
🔗	Download link for Debian 10 64 bit VmWare image	📄	✓		⋮
🔗	Article explaining sources.list and how to change VMWare image to download updates from the internet	📄	✓		⋮
			✓		⋮
			✓		⋮
			✓		⋮
			✓		⋮

Pro Apps are now Free for Personal Use and Licensed for Commercial Use

The most exciting part is that Fusion Pro and Workstation Pro will now have two license models. We now provide a Free Personal Use or a Paid Commercial Use subscription for our Pro apps. Users will decide based on their use case whether a commercial subscription is required.

This means that everyday users who want a virtual lab on their Mac, Windows or Linux computer can do so *for free* simply by registering and downloading the bits from the new download portal located at support.broadcom.com

VMware Fusion Pro Download

VMware Workstation Pro Download

Practical – Installing

- Download VMware 64 bit

Debian 10 Buster

VirtualBox
VMware
Info

Desktop Images

- VMware (VMDK) 32bit
Download
Size: 1.0GB
SHA256: 5e0effa84390d096674ef18527ce14900da3d8df4435da1d245e02922f9f45
- VMware (VMDK) 64bit
Download
Size: 995MB
SHA256: 76f033f673d00663a4181b10e53550797a7993c89f747fdbbd73e55f311e

Server Images

- VMware (VMDK) 32bit
Download
Size: 413MB
SHA256: BD6220B3E64C25D5AD88774738E03379BE2CE93AAC453845EECF5319870E88
- VMware (VMDK) 64bit
Download
Size: 465MB
SHA256: 24D3A401609670A5912DB14FF0A3B42FF9A9812BBC9154D7A6B459DA3AF156

Eksterne ressurser		
	Download link for VMWare (Workstation and Fusion)	
	Download link for Debian 10 64 bit VmWare image	
	Article explaining sources.list and how to change VMWare image to download updates from the internet	
	example_sources_nocdrom.txt	
	Article explaining how to install GCC on Debian 10	
	installing_gcc_with_output.txt	
	Article on how to use GDB to debug applications	

Practical – Installing

- Click on “Info” to find username, password, etc

Debian 10 Buster

VirtualBox

VMware

Info

Username: osboxes

Password: osboxes.org

Root Account Password: osboxes.org

VB Guest Additions & VMware Tools: Installed

Keyboard Layout: US (Qwerty)

VMware Compatibility: Version 10+

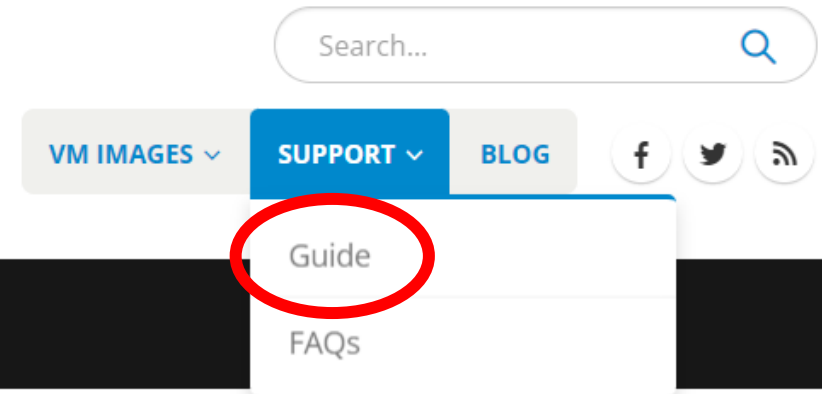
Eksterne ressurser		
	Download link for VMWare (Workstation and Fusion) 	
	Download link for Debian 10 64 bit VmWare image 	
	Article explaining sources.list and how to change VMWare image to download updates from the internet 	
	es_nocdrom.txt	
	How to install GCC on Debian 10 	
	with_output.txt	
	GDB to debug applications 	

Practical – Installing

- Before you try to install the image, first scroll to the top of the page and click on **SUPPORT** and choose **GUIDE**



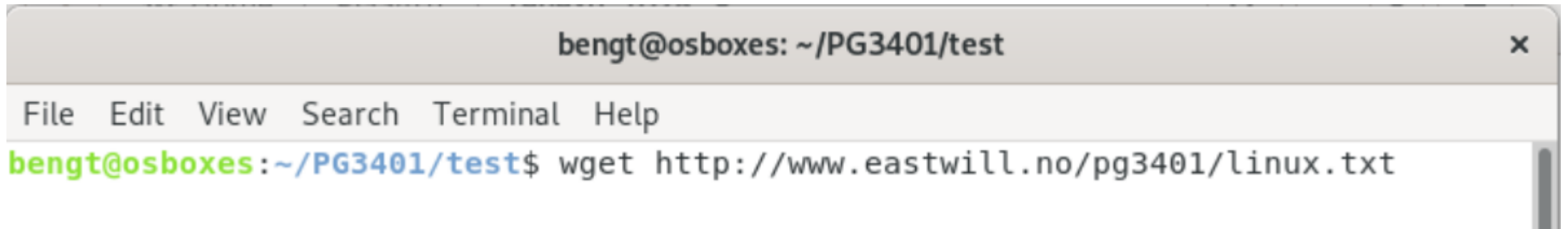
Debian



- Scroll down to “Vmware (WMDK file version)” and follow instructions

Practical – Linux

- To use “wget” to download a file, first open a terminal window by clicking on Activities in the top left corner – and type “Terminal”
- In the terminal window type the “wget” command



```
bengt@osboxes: ~/PG3401/test
```

File Edit View Search Terminal Help

```
bengt@osboxes:~/PG3401/test$ wget http://www.eastwill.no/pg3401/linux.txt
```

Practical – Linux

- Typing “ls” will show the file you just downloaded
- Typing “cat linux.txt” will print out it’s contents to the terminal
- Try to follow the instructions, using google if you need help

```
bengt@osboxes: ~/PG3401/test
File Edit View Search Terminal Help
bengt@osboxes:~/PG3401/test$ wget http://www.eastwill.no/pg3401/linux.txt
--2023-01-10 16:15:48-- http://www.eastwill.no/pg3401/linux.txt
Resolving www.eastwill.no (www.eastwill.no)... 46.30.215.243, 2a02:2350:5:107:80
9e:73cc:fc44:5e87
Connecting to www.eastwill.no (www.eastwill.no)|46.30.215.243|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2094 (2.0K) [text/plain]
Saving to: 'linux.txt.1'

linux.txt.1      100%[=====>]    2.04K  --.-KB/s    in 0s

2023-01-10 16:15:48 (108 MB/s) - 'linux.txt.1' saved [2094/2094]

bengt@osboxes:~/PG3401/test$ ls
linux.txt  linux.txt.1
bengt@osboxes:~/PG3401/test$ cat linux.txt
Good job students!

Now you have learnt how to get files over the internet with URL.

Practical Session :

Goals:
```

Lab exercises

- **Help each other becoming used to Linux. Some of you might have used it before, some are new to this OS. Make sure you all help each other – this is one of the most difficult courses you will take at this school, everyone of you will need help from the class! :-) You must attend Campus trainings, trying to learn it all your self will most often fail, hard...**
- **After next week everyone in the class MUST have a running Linux environment and be able to edit and compile code!**
- **NOTE: Do not copy/paste source code from slides, Powerpoint will have formatted a double-quote (") and other characters (for instance -), and it is almost guaranteed to insert extra spaces when you copy.**
- **Instead WRITE the source code you see on the slides into your editor...**

Lab exercises

Next week we will install GCC, a compiler for Linux, and we will learn more about how to use Linux – so what we have done is enough for today 😊

It is REQUIRED to use Debian 10 64 bit and GCC, earlier classes have had many students trying to use other tools and other Linux distros (including just compiling on Mac), and some ended up with very strange problems I had problems solving...

This course is named “C Programming for Linux”.



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